



UNITED INTERNATIONAL UNIVERSITY

Dept : Computer Science and Engineering

Course Name: Physics

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Section : B

Assignment Number: 2

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Ans. to the Q.NO - (7)

We know, r = damping factor, which is used in DHM, So x vs t graph will be of DHM.

We know, there are 3 types of x vs t graph in DHM—

(i) oscillatory : $\omega^2 > \frac{r^2}{4}$

(ii) critical damping : $\omega^2 = \frac{r^2}{4}$

(iii) Over damping : $\omega^2 < \frac{r^2}{4}$

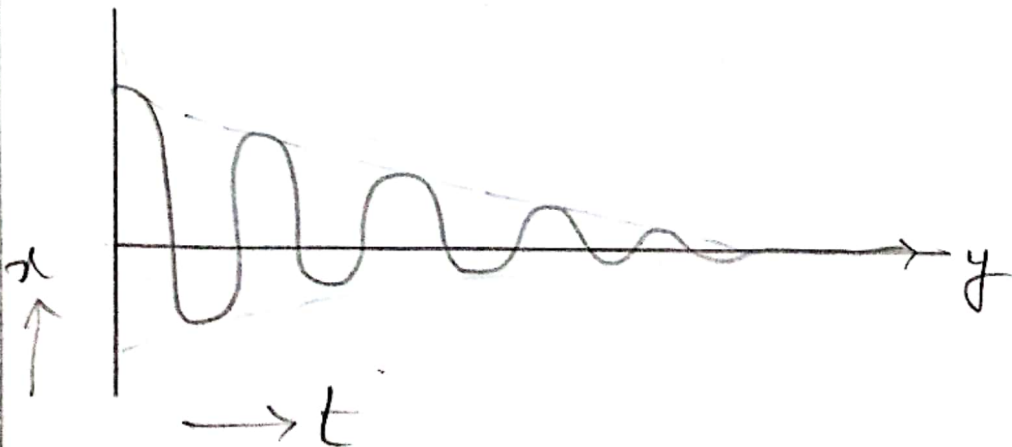
(i) given, $\frac{\omega}{r} = 10$

$$\Rightarrow \omega = 10r$$

$$\Rightarrow \omega = \frac{r}{2} \times 20$$

$\therefore \omega > \frac{r}{2}$ here.

$\therefore$ the graph will show oscillatory behaviour here.



(ii) given, $\frac{\omega}{\gamma} = 0.5$

$$\Rightarrow \omega = 0.5\gamma$$

$$\Rightarrow \omega = \frac{\gamma}{2}$$

$$\therefore \omega = \frac{\gamma}{2}$$

$\therefore$ this graph will show critical damping.



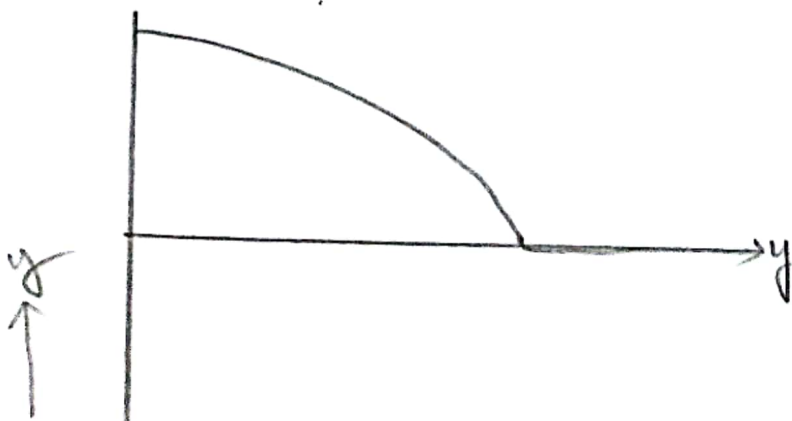
(iii) given, $\frac{\omega}{\gamma} = 0.03$

$$\Rightarrow \omega = 0.03\gamma$$

$$\Rightarrow \omega = \frac{\gamma}{2} \times 0.06$$

$$\therefore \omega < \frac{\gamma}{2}$$

$\therefore$ this graph will show over damping.



Ans. to the Q. NO (-2)

We know,

$$\begin{aligned}\text{For DMM, } x(t) &= Ae^{-\alpha t} \cos(\omega t + \phi) \\ &= 7e^{-\alpha t} \cos(6t + \phi)\end{aligned}$$

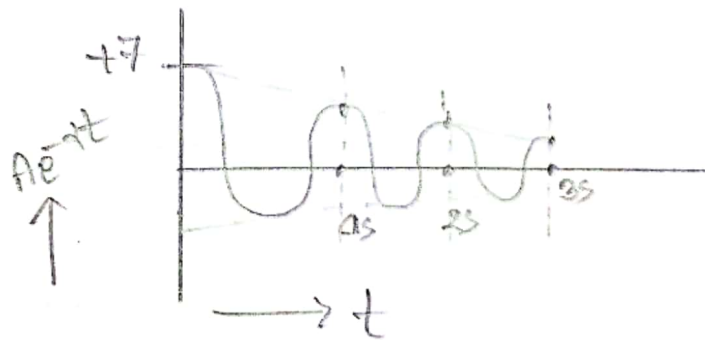
So, we will draw the graph for ~~everyone~~ DMM.

Here, Amplitude, $A = 7$ m

Angular frequency, $\omega = 6$ rad/s

Number of cycle = 3

$$\text{Time Period, } T = \frac{2\pi}{\omega} = \frac{2\pi}{6} = 1.05 \approx 1\text{s}$$



Ans. to the Q. NO - (3)

Inductance, $L = 0.4 \text{ H}$

Capacitance, $C = 0.0020 \text{ MF} = 0.0020 \times 10^{-6} \text{ F}$

(i) We know, in RLC circuit, the condition for behaving to be oscillatory is $\frac{1}{LC} > \frac{R^2}{4L^2}$

$$\therefore \frac{1}{LC} = \frac{R^2}{4L^2}$$

$$\Rightarrow LCR^2 = 4L^2$$

$$\Rightarrow R^2 = \frac{4L^2}{LC} = \frac{4L}{C}$$

$$\Rightarrow R = \sqrt{\frac{4L}{C}}$$

$$\Rightarrow R = 2 \sqrt{\frac{L}{C}}$$

$$= 2 \times \sqrt{\frac{0.4}{0.0020 \times 10^{-6}}}$$

$$= 28284.27125 \Omega$$

(ii)

Resonant frequency, $f = \frac{1}{2\pi \sqrt{LC}}$

$$= \frac{1}{2\pi \sqrt{0.4 \times 0.0020 \times 10^{-6}}}$$

$$= 5626.98 \text{ Hz}$$

Ans

Ans. to the Q. NO - (4)

Capacitance, $C = 0.1 \mu F = 0.1 \times 10^{-6} F$

Inductance, $L = 10 mH = 10 \times 10^{-3} H$

Resistance, $R = 200 \Omega$

the condition to be oscillatory is: $\frac{1}{LC} > \frac{R^2}{4L^2}$

$$\therefore \frac{1}{LC} = \frac{1}{(10 \times 10^{-3}) \times (0.1 \times 10^{-6})} = 1 \times 10^9$$

$$\therefore \frac{R^2}{4L^2} = \frac{(200)^2}{4 \times (10 \times 10^{-3})} = 1 \times 10^6$$

$$\therefore \frac{1}{LC} > \frac{R^2}{4L^2}$$

$\therefore$ the circuit is oscillatory.

$$\begin{aligned} \therefore \text{frequency of oscillation, } f &= \frac{1}{2\pi} \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}} \\ &= \frac{1}{2\pi} \sqrt{1 \times 10^9 - 1 \times 10^6} \\ &= 5 \times 10^3 \end{aligned}$$

$$\begin{aligned} \therefore \text{Resonant frequency, } f &= \frac{1}{2\pi \sqrt{LC}} \\ &= \frac{1}{2\pi \sqrt{}} \\ &= 5 \times 10^3 \end{aligned}$$

Ans. to the Q.NO - (5)

$$m = 250g = 0.25kg$$

$$K = 85 \text{ Nm}^{-1}$$

$$b = 70 = 0.07$$

$$\omega = \sqrt{\frac{K}{m}} = \sqrt{\frac{85}{0.25}} = 18.44 \text{ rad s}^{-1}$$

$$(i) \quad T = 2\pi \sqrt{\frac{m}{K}} = 2\pi \sqrt{\frac{0.25}{85}} = 0.34s$$

$$(ii) \quad A(t) = \frac{A_0}{2}$$

$$\Rightarrow A_0 e^{-\alpha t} = \frac{A_0}{2}$$

$$\Rightarrow e^{-\alpha t} = \frac{1}{2}$$

$$\Rightarrow \ln e^{-\alpha t} = \ln\left(\frac{1}{2}\right)$$

$$\Rightarrow -\alpha t = \ln\left(\frac{1}{2}\right)$$

$$\Rightarrow \alpha t = 6.9 \times 10^{-1}$$

$$\Rightarrow t = \frac{6.9 \times 10^{-1}}{0.14}$$

$$\alpha = \frac{b}{2m} = \frac{0.07}{2 \times 0.25} = 0.14$$

$$\therefore t = 4.93s$$

(iii) number of Oscillations, $n = \frac{\tau}{T} = \frac{3.57}{0.341} = 10.46$

(iv) life time, $\tau = \frac{1}{\gamma} = \frac{m}{b} = \frac{0.25}{0.07} = 3.57 \text{ s}$

(v) Damping energy, $E = (E_0)e^{-2\alpha t}$

$$= \left(\frac{1}{2} K A^2\right) e^{-2\alpha t}$$

$$= \left(\frac{1}{2} 85 \times 35^2\right) e^{-2\alpha t}$$

$$= 52062.5 \times e^{-2(0.14) \times 10.46}$$

$$= 2783.32 \text{ J}$$

$A = 35$
 $K = \omega^2 m = (18.44)^2 \times 0.25 = 85$
 $\alpha = \frac{b}{2m} = 0.14$
 $t = 10.46$
 $n = \frac{\tau}{T}$

Ans

(vi)

$$\frac{A_d}{A_0} = \frac{A_0 e^{-\alpha t}}{35}$$

Ans. to the Q. NO - (6)

We know, $y = A \sin \frac{2\pi}{\lambda} (vt - x)$

$$= 4 \sin (0.10x - 2t)$$
$$= 4 \sin (-2t + 0.10x)$$
$$= 4 \sin \{-(2t - 0.10x)\}$$
$$= 4 \sin \left\{ -x \cdot 0.10 \left(\frac{2t}{0.10} - x \right) \right\}$$

#

(i) Amplitude, $A = 4 \text{ m}$

(ii) ~~$v = f\lambda$~~ here, $\frac{2\pi}{\lambda} = -0.10$

$$\Rightarrow 2\pi = -0.10\lambda$$
$$\Rightarrow -62.83 \text{ m} = \lambda$$

(iii) ~~$v = \frac{2\pi}{0.10}$~~

$$\therefore v = 20 \text{ m s}^{-1}$$

(iv) $f = \frac{v}{\lambda} = \frac{20}{-62.83} = -0.318 \text{ Hz}$

Ans. to the Q. NO (7)

$$y = A \sin \frac{2\pi}{\lambda} (vt - x) = \cancel{4} \sin 2\pi$$
$$= 4 \sin \left(\frac{2\pi vt}{\lambda} - \frac{2\pi x}{\lambda} \right)$$

when $t=0$ $y = A \sin \left(0 - \frac{2\pi x}{\lambda} \right)$

$$= A \sin -\frac{2\pi x}{\lambda}$$
$$= -A \sin \frac{2\pi x}{\lambda}$$
$$= -4 \sin \frac{2\pi x}{100}$$

$$\therefore A = \cancel{-4} \text{ m}$$

$$\therefore \lambda = 100 \text{ m}$$

given $v = 30 \text{ m/s}$

$$\therefore y(t=3s) = -4 \sin \frac{2\pi}{\lambda} (3 \times 30 - x)$$
$$= -4 \sin \frac{2\pi}{100} (90 - x)$$
$$= -4 \sin \frac{\pi}{50} (90 - x)$$

Ans

Ans. to Que. Q.NO-8

$$\begin{aligned} y &= A \sin \frac{2\pi}{\lambda} (vt - x) \\ &= 5 \sin \pi (360t - 0.15x) \\ &= 5 \sin \left\{ \pi \times 0.15 \left(\frac{360t}{0.15} - x \right) \right\} \end{aligned}$$

(i) $A = 5 \text{ m}$

(ii) $\frac{2\pi}{\lambda} = 0.15\pi$

$$\Rightarrow \frac{2}{\lambda} = 0.15$$

$$\Rightarrow 0.15\lambda = 2$$

$$\Rightarrow \lambda = 13.33 \text{ m}$$

(iii) $vt = \frac{360t}{0.15}$

$$\Rightarrow v = 2400 \text{ m/s}$$

(iv) $v = f\lambda$

$$\therefore f = \frac{v}{\lambda} = \frac{2400}{13.33} = 180 \text{ Hz}$$

(v) $T = \frac{1}{f} = \frac{1}{180} = 0.0055$

Ans. to the Q. No - (9)

$$y = A \sin \frac{2\pi}{\lambda} (vt - x)$$

$$\Rightarrow \frac{y}{A} = \sin 2\pi \left(\frac{t}{T} - \frac{x}{\lambda} \right)$$

in first case,

$$\frac{y_1}{A} = \sin 2\pi \left(\frac{t}{T} - \frac{x_1}{\lambda} \right)$$

here, $y_1 = 6$, $A = 8$, $x_1 = 10$ cm

$$\frac{6}{8} = \sin 2\pi \left(\frac{t}{T} - \frac{10}{\lambda} \right) \quad \text{--- (1)}$$

$$\frac{4}{8} = \sin 2\pi \left(\frac{t}{T} - \frac{20}{\lambda} \right) \quad \text{--- (11)}$$

$$\text{from (1)} : \Rightarrow 2\pi \left(\frac{t}{T} - \frac{10}{\lambda} \right) = \sin^{-1} \left(\frac{6}{8} \right)$$

$$\Rightarrow \frac{t}{T} - \frac{10}{\lambda} = 0.14 \quad \text{--- (111)}$$

$$\text{from (11)} : \Rightarrow 2\pi \left(\frac{t}{T} - \frac{20}{\lambda} \right) = \sin^{-1} \left(\frac{4}{8} \right)$$

$$\Rightarrow \frac{t}{T} - \frac{20}{\lambda} = 0.08 \quad \text{--- (112)}$$

$$\therefore \frac{15}{\lambda} = 0.06$$

$$\Rightarrow \lambda = 250 \text{ cm}$$

Ans

Ans. to the Q.No - (10)

We know, for simple pendulum, $T = 2\pi \times \sqrt{\frac{L}{g}}$

$$\Rightarrow \frac{T}{2\pi} = \sqrt{\frac{L}{g}}$$

$$\left| \begin{array}{l} \text{length, } L = ? \\ T = 5 \text{ s} \\ g = 9.8 \text{ m/s}^2 \end{array} \right|$$

$$\Rightarrow \left(\frac{T}{2\pi} \right)^2 = \frac{L}{g}$$

$$\Rightarrow \frac{5^2}{4\pi^2} = \frac{L}{g}$$

$$\Rightarrow 0.633 = \frac{L}{g}$$

$$\Rightarrow L = 0.633 \times 9.8$$

$$\Rightarrow L = 6.21 \text{ m}$$

$\therefore$ length 6.21 m

$\therefore$ frequency, $f = \frac{1}{T}$

$$= \frac{1}{5}$$

$$= 0.2 \text{ Hz.}$$

Ans

Ans to the Q.No (13)

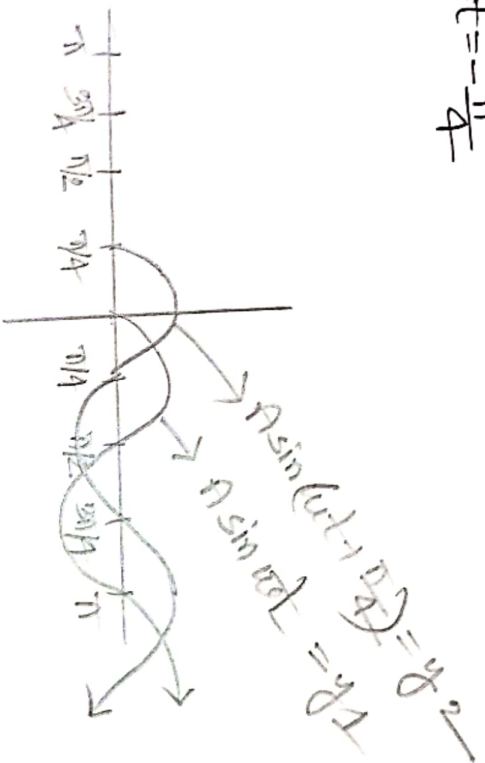
①

$$y_1 = A \sin \omega t = A \sin (\omega t + 0)$$

$$y_2 = A \sin (\omega t + \frac{\pi}{4})$$

$$\therefore \omega t = 0$$

$$\omega t = -\frac{\pi}{4}$$



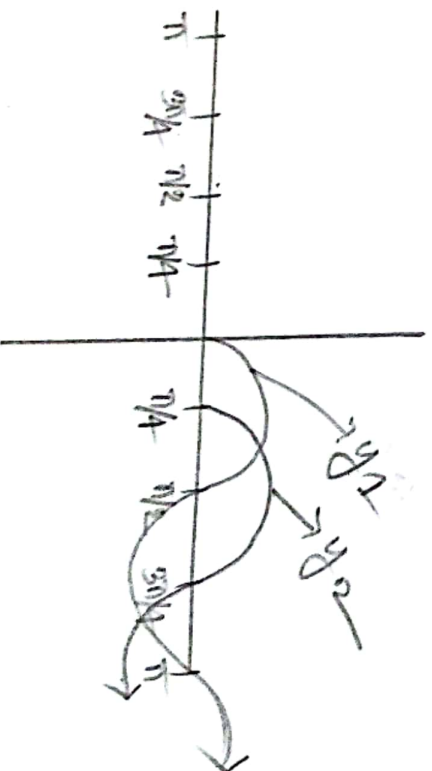
②

$$y_1 = A \sin \omega t = A \sin (\omega t + 0)$$

$$y_2 = A \sin (\omega t + \frac{\pi}{4})$$

$$\therefore \omega t = 0$$

$$\omega t = \frac{\pi}{4}$$



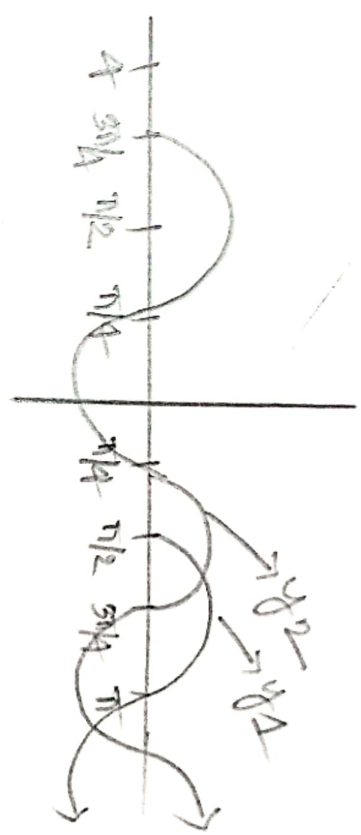
(III)

$$y_1 = A \sin \left(\omega t - \frac{\pi}{2} \right)$$

$$y_2 = A \sin \left(\omega t + \frac{3\pi}{4} \right)$$

$$\therefore \omega t = \frac{\pi}{2}$$

$$\omega t = \frac{1}{2} \frac{3\pi}{4}$$



(IV) $y = A \sin (\omega t + \phi)$

$$\phi = 45^\circ = \frac{\pi}{4}$$

$$\phi = -\frac{\pi}{2}$$

$$\therefore \omega t + \phi = 0$$

$$\Rightarrow \omega t + \frac{\pi}{4} = 0$$

$$\Rightarrow \omega t = -\frac{\pi}{4} \quad \textcircled{y_1}$$

$$\omega t + \left(-\frac{\pi}{2} \right) = 0$$

$$\Rightarrow \omega t - \frac{\pi}{2} = 0$$

$$\Rightarrow \omega t = \frac{\pi}{2} \quad \textcircled{y_2}$$

